



lecture 11

In the previous class we showed that

$$G(s) = K \frac{\prod_{i=1}^m (s - z_i)}{\prod_{i=1}^n (s - p_i)}$$

We note that $G(s)$ is a rational function and it can be expressed as sum of simpler terms, and its Laplace transform can be found in the table.

We assume that the poles are real or complex but distinct.

We can rewrite $G(s)$ as partial fraction expansion.

$$G(s) = \frac{C_1}{s-p_1} + \frac{C_2}{s-p_2} + \dots + \frac{C_n}{s-p_n}$$

next we determine the set of constants $\{C_i\}$

Once the value of C_i 's are determined, we can take the inverse Laplace transform, and the time function becomes

$$g(t) = \sum_{i=1}^n C_i e^{p_i t} 1(t), \quad t \geq 0$$

$$\text{If } F(s) = \frac{1}{s-p_i}$$

$$f(t) = e^{p_i t} 1(t)$$

